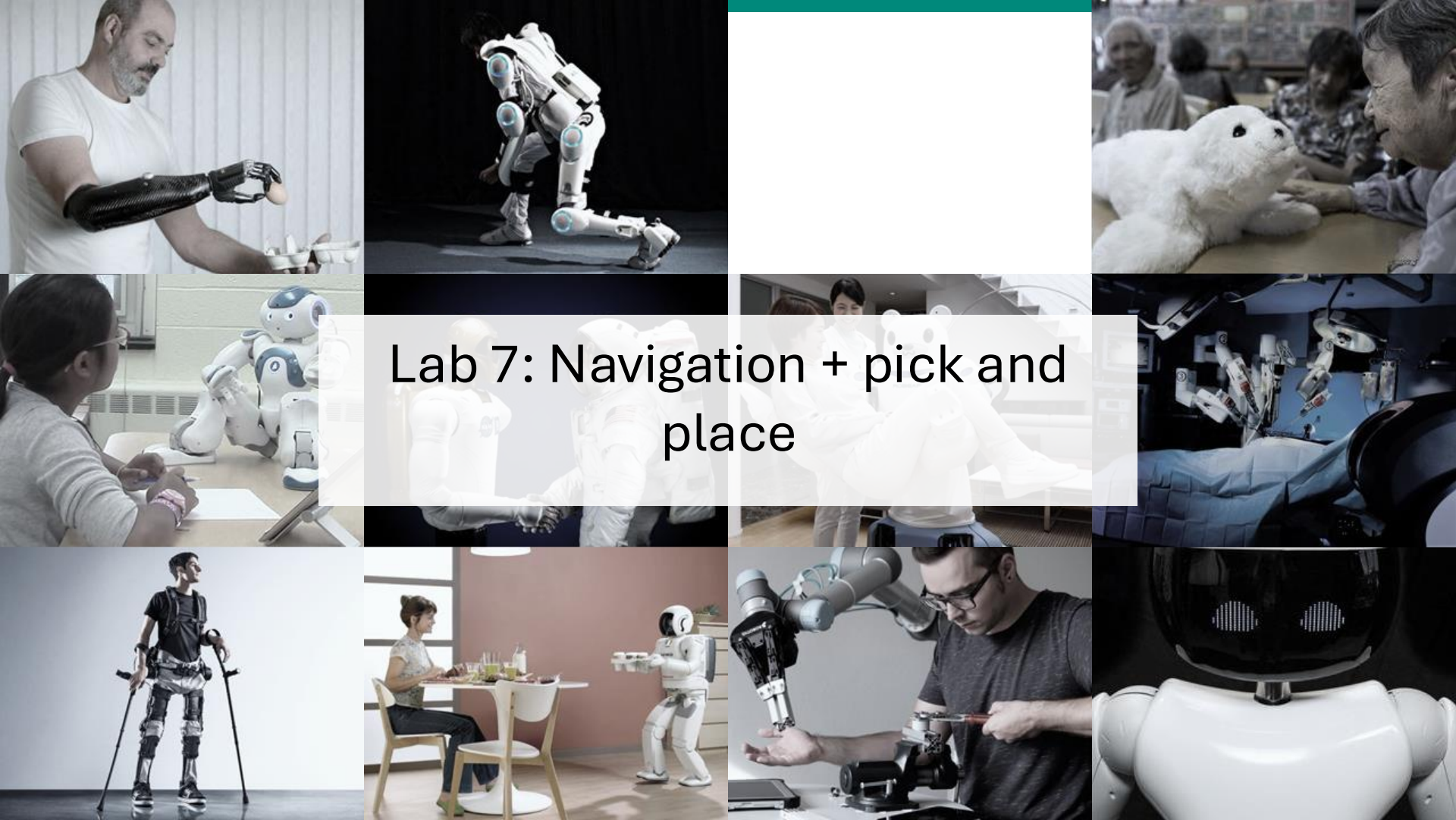

Lab 7: Navigation + pick and place

This Is an Integration Lab

- You have all the building blocks from:
- **Lab 5: Potential field navigation with obstacle avoidance**
- **Lab 6: Gradient descent IK for manipulation**

- **Your challenge: Combine them on the PR2 robot**

Overview

- 1. Navigate through the environment
- 2. Pick up the objects
- 3. Place them at the place zone

The PR2 Platform

- **Mobile base**
- Differential drive (like TIAGo)
- 360° lidar for obstacle detection
-
- **5-DOF right arm**
- Gradient descent IK (like UR5e, but 5 joints)
- Parallel jaw gripper
- Adjustable torso height

Collision avoidance

- **Note: -5 points/collision**
- Implement the collision avoidance from lab 5 or add your own!

Tips for Success

- **Build and test incrementally**
- FK → Jacobian → IK → Navigation → Full integration
-
- **Tune your gains carefully**
- REPULSE_GAIN balance: avoid oscillation AND collisions
-
- **Watch your coordinate frames**
- Use world_to_base() for IK targets

Grading and submission

Code (FK, Jacobian, IK, Navigation, Pick, Place)	**70 pts**
Lab Report (FK verification, IK analysis, collision analysis)	**30 pts**
Collision Penalty	**-5 pts each**

Questions?